

ANANTA TRADING ENGINE

Comprehensive Technical Dossier & Quantitative Audit Package

Prepared for independent quantitative research review. This documentation is written to be sufficient for an experienced systematic trader / researcher to understand, audit and recommend improvements to the engine **without access to the source code**.

Grounded in Lab run	a7a8a0c0-2080-4405-97e7-f97c36cc2f76
Symbols	BTC/USD, SOL/USD, ETH/USD
Period / timeframe	1y (2025-07-20 -> 2026-07-15) · 1h (live-execution baseline); 15m & 30m also replayed for com
Exit method	ATR Exit (x2.5 stop, 14p, arm 3%, trail x2)
Report generated	2026-07-16T13:05:16 (source PDF)
Dossier generated	2026-07-16 16:42 UTC
Companion artifacts	trade-log CSV · per-candle indicators CSV · config JSON

Provenance & honesty note. Every performance figure in this dossier is transcribed verbatim from the supplied backtest report (run a7a8a0c0-2080-4405-97e7-f97c36cc2f76). Where a metric the review asks for is **not present in that run**, it is explicitly flagged as UNAVAILABLE rather than estimated. Engine mechanics are described from the actual source code.

Part 1 — Project Overview

What Ananta is. Ananta is a single-operator algorithmic trading operating system for spot crypto. It researches, validates, paper-trades and (optionally) live-executes a small set of independent technical alpha models, wrapped in a deterministic risk layer and an AI copilot for analysis and reporting.

Primary goal. Trade with evidence, not assumptions: no strategy reaches real capital until it has been validated offline (Research Lab) and paper-traded. The engine is deliberately technical-first.

Core philosophy. (1) Entries and exits are **pure, deterministic compute** — zero LLM involvement, so behaviour is reproducible and auditable. (2) Independent alpha models with distinct 'personalities' rather than one monolithic strategy. (3) Regime-gating decides which model may even act. (4) Risk is owned by a separate, priority-arbitrated exit engine, not by the entry logic.

Aspect	Detail (this build)
Supported exchange	Kraken (LIVE via CCXT); Coinbase selectable
Assets (this run)	BTC/USD, SOL/USD, ETH/USD (spot)
Timeframes	1h live-execution baseline; 15m / 30m for comparison; 4h/1d used internally for regime & zones
Order types	Market/limit-style fills; PAPER post-only maker simulation; LIVE limit with slippage cap
Workflows	Research Lab (offline) → Paper trading → Live (interlocked)
AI components	Macro-context (Gemini), Ask-Ananta copilot, weekly review — ADVISORY only, never trade-gating
Risk philosophy	Deterministic kill-switches + priority-arbitrated Universal Exit Engine

High-level architecture (data → decision → execution).

Market data (CCXT/Kraken OHLCV + orderbook) → indicator & regime layer → regime router (which model may act) → independent alpha models (Hunter / Squeeze / Continuation) produce a technical entry signal → risk fusion & kill-switches → position sizing → order execution (paper/live) → position watcher + Universal Exit Engine (modules A–F) → trade ledger & analytics → AI reporting. A separate Research Lab replays the identical live functions over history for validation.

Part 2 — Trading Engine Architecture

The live loop runs on a fixed cadence (default 90s trading loop; position watcher on its own interval). The offline Lab replays the **exact same functions** (`classify_regime`, `router`, `evaluate_primary/squeeze/continuation`, `evaluate_exit_engine`) bar-by-bar — only the clock and data source differ, giving live/backtest parity.

What happens on every new candle:

- **Data ingestion** — pull/append OHLCV (1h execution series; 4h/1d for regime & support zones) and an orderbook snapshot (spread + imbalance) for the risk gate.
- **Indicators** — EMA20/50/200, RSI14, MACD, ATR14, ADX14, Bollinger bandwidth are computed causally on the trailing analysis window (750 bars, matching live `EXEC_BARS_LIMIT`).
- **Regime** — `classify_regime()` labels the market (`TREND_UP` / `TREND_DOWN` / `RANGE` / `COMPRESSION` / `REVERSAL` / `NEUTRAL`).
- **Routing** — the regime decides which alpha model(s) are ALLOWED to evaluate (`hunter_allowed` / `squeeze_allowed` / `continuation_allowed`).
- **Signal generation** — the first allowed model that fires wins; it returns an entry profile + structural stop + confidence.
- **Validation / risk fusion** — kill-switches (spread, daily-loss, manual) and a tri-state circuit breaker; only an existential VETO can block. Macro/news/sentiment can NEVER create or block an entry.
- **Position sizing** — fixed USD lot (this run: \$75) capped at 95% of cash, or legacy %-of-equity scaled by confidence.
- **Execution** — entry fills at next-bar OPEN in the backtest (no look-ahead); taker fee + 0.05%/leg slippage on every leg.
- **Monitoring & exit** — the position watcher runs the Universal Exit Engine (modules A–F) each tick; a pessimistic intrabar LOW pass catches stops/trails first, then a CLOSE pass handles upside modules.
- **Performance tracking** — the closing leg records MFE/MAE, exit module, best/worst achievable exit, hold time, trade-quality score into the ledger for analytics.

Part 3 — Strategy Framework

Ananta ships **three core alpha models** (deterministic, hand-built) plus a **declarative catalog** of eight indicator strategies. All eleven were selected in this run's configuration.

IMPORTANT AUDIT CAVEAT — The offline Lab backtester's entry scan (lab/backtest.py:: _scan_entry) wires **ONLY** the three core models (hunter, squeeze, continuation). The eight declarative strategies are registered, tunable and run in the live/paper engine, but they produced **ZERO** entries in this backtest. Consequently **ALL 673 trades** in the supplied report originate from the three core models. This is confirmed by the regime breakdowns, whose trade counts sum exactly to the per-symbol totals (BTC 10+70+1+8+108=197, etc.).

Core model — HUNTER (reversal · 'buys fear')

- **Objective / thesis:** buy validated historical support after a capitulation, not a breakout.
- **Ideal:** deep oversold pullbacks into 1–2y support. **Worst:** trending-down knife-catches without a reset.
- **Gates (ALL required):** historical support zone · pullback confirmation (no green-candle chase) · volume exhaustion (falling volume slope) · 4H RSI(14) momentum reset $\leq$ rsi_reset_max.
- **Entry profiles:** AGGRESSIVE_PULLBACK / STABILIZED_REVERSAL / DEEP_DISCOUNT.
- **Structural stop:** 0.4×ATR below the structure low. Rejection reasons are logged (no-zone, chasing, volume-not-exhausted, rsi-not-reset).

Core model — SQUEEZE (volatility expansion · 'buys expansion')

- **Trigger:** Bollinger Bands fully inside Keltner Channels (volatility coiled).
- **Entry:** NOT the first breakout candle — either CONTINUATION (breakout → inside bar → break of its high) or RETEST (breakout → pull back to the 20-MA basis → reclaim).
- **Stop:** hard stop at the 20-period MA (Bollinger basis). ATR trail thereafter.

Core model — CONTINUATION (trend pullback · 'buys the dip in an uptrend')

- **Trigger:** established uptrend — 50-EMA rising AND 20-EMA > 50-EMA AND price above the 50-EMA.
- **Entry:** controlled pullback to the 20-EMA with volume drying up and RSI in a healthy 40–62 band (NOT the 30–35 reversal zone), then a stabilising candle.
- **Structural stop:** below the pullback low / 50-EMA, buffered by 0.4×ATR.

Declarative catalog (registered & tunable; not active in this run's entries)

Key	Name	Entry logic (long-only)	Exit logic	Conf.
ema-cross	EMA Cross	Fast EMA crossed above slow EMA	ema_fast cross_below ema_slow	63
supertrend	Supertrend	Supertrend flipped bullish	st_dir cross_below 0	71
rsi-momentum	RSI Momentum	RSI momentum breakout above trend EMA	It \$rsi_exit	64
macd-trend	MACD Trend	MACD crossed above signal in bullish regime	macd cross_below sig	61
bollinger-mr	Bollinger Mean Reversion	Price closed below the lower Bollinger band	close gte mid	72
donchian-breakout	Donchian Breakout	Price broke above the Donchian channel high	close lt lo	68
atr-breakout	ATR Breakout	Price broke previous close + k×ATR	(exit via risk engine)	65
keltner-breakout	Keltner Breakout	Price closed above the upper Keltner band	close cross_below mid	67

Each declarative strategy exposes tunable parameters (periods, multipliers, thresholds) — see the config JSON export for exact defaults, bounds and the full boolean spec (indicators + entry/exit conditions).

Pseudocode (representative — EMA Cross): if ema(fast) crosses_above ema(slow): enter_long(); if ema(fast) crosses_below ema(slow): exit(). Every catalog strategy compiles to this same declarative form (indicators → entry conditions → exit conditions).

Part 4 — Entry Decision Logic

Entries are **technical-first and conjunctive**: a core model fires only when ALL of its gates are simultaneously true (boolean AND), after the regime router has admitted that model. There is no weighted score that can rescue a failed gate; confidence is recorded for research but does not lower a gate. Priority order when multiple models are allowed: Hunter → Squeeze → Continuation (first trigger wins, one position at a time).

- **Regime filter** (router): e.g. Hunter is admitted in reversal/neutral regimes, Squeeze around compression→expansion, Continuation in established uptrends.
- **Trend filter**: EMA stack (20/50/200) + optional 4H HTF alignment (price>EMA50>EMA200).
- **Momentum filter**: RSI reset (Hunter) or RSI band (Continuation) or RSI breakout (declarative rsi-momentum).
- **Volatility filter**: ATR percentile & Bollinger-in-Keltner coil (Squeeze) / ATR expansion (declarative).
- **Volume filter**: falling volume slope into support (Hunter) / volume spike on break (Squeeze).
- **Support/Resistance**: multi-year daily/4H horizontal zones with touch-count & strength (Hunter).
- **Multi-timeframe**: 4h/1d used for regime & zones; 1h for execution.
- **AI contribution**: NONE at the entry gate. Macro bias/confidence is advisory context only (see Part 14).

Decision tree (core): regime admits model? → support/trend context valid? → pullback/coil confirmed (no chase)? → momentum reset/band ok? → volume ok? → risk gate (spread/daily-loss/veto) clear? → BUY with structural stop; else HOLD with an explicit reason code.

Part 5 — Exit Decision Logic

Exits are owned by the **Universal Exit Engine**: every module is evaluated independently each tick and a single-pass priority arbitration executes the highest-priority firing module.

Module	Prio	Action	Trigger
A Structural failure	1	EXIT_FULL	structural stop / % stop / locked profit-floor breach
KILL Emergency	2	EXIT_FULL	existential circuit-breaker veto (hack/insolvency/reg)
F Profit protection	3	TIGHTEN	breakeven at +1R, then lock a +1% profit floor
B Momentum exhaustion	4	EXIT 50%	overbought zone + volume climax + exhaustion candle
S Structure failure	5	EXIT_FULL	lower-low AND momentum dead (RSI<50 & below 20-EMA)
D EMA trend loss	5	EXIT_FULL	close below 20-EMA / dead-cross
C ATR trail	6	EXIT_FULL	armed trailing stop = peak – X·ATR (arms at +2R)
E Time exit	7	EXIT_FULL	capital-efficiency time cap per strategy profile

This run used the pure ATR exit (not the full native engine): initial stop = entry – 2.5×ATR(14); once MFE ≥ 3% the stop trails at peak – 2.0×ATR and only ever tightens; exit when a bar LOW breaches the active stop. Positions still open at window end are marked out at the last close (labelled EOD). The Lab also A/B/C-tests Fixed-\$ exits on the SAME entries (Part 10/11 tables).

Part 6 — Risk Management System

- **Hard kill-switches** (block ALL new entries / force exit): manual kill, spread breach ($\text{spread} > \text{max_spread_pct} = 0.5\%$), daily-loss breach ($\text{day P\&L} \leq -\text{max_daily_loss_pct} = -10.0\%$).
- **Confidence gate**: $\text{min_confidence} = 0.8$ (blocks new trades, not a hard stop).
- **Max concurrent positions**: 8 (bounds = 1–20).
- **Position sizing**: fixed USD lot (normal \$75.0 / strong \$75.0; this run \$75) capped at 95% of cash; legacy path scales 1.0–3.0% of equity by confidence.
- **Structural stop**: per-trade, set by the model ($0.4 \times \text{ATR}$ below structure); default $\text{stop_loss_pct} = 10.0\%$.
- **Circuit breaker**: tri-state PASS / CAUTION / VETO; only existential VETO acts (sentiment/macro can never reach VETO).
- **Fees/slippage**: taker 0.4% / maker 0.25%; backtest applies 0.05% slippage per leg.

GAPS an auditor should note: no explicit portfolio-level max-exposure cap beyond position count $\times$ lot; no correlation cap across BTC/ETH/SOL (highly correlated) — three independent \$1,200 books were run in parallel; no leverage (spot only); no volatility-scaled position sizing in this run (flat \$75).

Full numeric bounds for every tunable field are in the config JSON (`risk_settings_hard_bounds`).

Part 7 — Backtest Configuration (this run)

Parameter	Value
Run ID	a7a8a0c0-2080-4405-97e7-f97c36cc2f76
Kind	backtest (offline historical replay)
Symbols	BTC/USD, SOL/USD, ETH/USD
Period	1y (2025-07-20 -> 2026-07-15)
Execution timeframe	1h (15m & 30m replayed for comparison — both produced 0 trades)
Exit method	ATR Exit (x2.5 stop, 14p, arm 3%, trail x2)
Position size	\$75 flat notional per trade
Baseline capital	\$1,200 per symbol (independent books)
Selection metric	return_over_dd
Commission	RiskSettings.taker_fee_pct = 0.4% per leg
Slippage	0.05% per leg (synthetic)
Entry fill	next-bar OPEN (signal on closed bar i → fill i+1)
Exit fill	pessimistic intrabar LOW pass then CLOSE pass
Warm-up	200 bars before the window (EMA200 / regime)
Analysis lookback	750 trailing bars fed to strategy fns (live parity)
Look-ahead prevention	point-in-time zones; closed-bar signals; next-bar fills
Compounding	realised P&L accrues to the per-symbol book (equity curve)
Missing data / survivorship	single continuous Kraken OHLCV series per symbol (no delisting universe)
Randomisation	none — fully deterministic

Part 8 — Complete Parameter Configuration

The full, machine-readable parameter set — every RiskSettings default and hard bound, ATR-exit params, exit-comparison grid, regime thresholds, entry-quality components, and every declarative strategy's tunables (EMA/RSI/MACD/ATR/Bollinger/Donchian/Keltner periods & multipliers) with defaults and min/max — is provided in the companion file **ananta_engine_config.json**. Key live defaults:

Field	Default	Field	Default
max_spread_pct	0.5	max_daily_loss_pct	10.0
min_confidence	0.8	max_concurrent_positions	8
normal_lot_usd	75.0	stop_loss_pct	10.0
trail_arm_pct	5.0	trail_distance_pct	3.0
taker_fee_pct	0.4	maker_fee_pct	0.25
rsi_reset_max	35.0	squeeze_vol_expansion_min	1.5

Part 9 — Trade Lifecycle (worked example)

Representative closed trade from the supplied report — BTC/USD trade #22:

Stage	Detail
Market data	1h BTC/USD candles; regime + support zones from 4h/1d
Signal	core-model trigger on closed bar (entry gates all true)
Entry fill	2025-10-01 13:00 UTC @ \$116,332.93 (next-bar open + slippage)
Position size	\$75 notional → 0.0006447 BTC
Monitoring	ATR stop = entry – 2.5×ATR; trails at peak – 2.0×ATR once MFE ≥ 3%
Exit	2025-10-02 05:00 UTC @ \$118,381.83, module ATR
Result	+\$0.72 net (after fees + slippage)

Every trade in the run is provided row-by-row in `ananta_trade_log.csv` (parsed directly from the report): trade #, entry/exit UTC, entry/exit price, size, net P&L, exit module.

UNAVAILABLE in the supplied report's trade table (and therefore not in the CSV): per-trade strategy attribution, entry profile, regime-at-entry, MFE/MAE, and indicator values at entry/exit. These fields ARE captured by the engine internally (`lab/backtest.py` trade dicts) but were not rendered in the PDF. The per-candle indicator CSV lets a reviewer reconstruct the indicator state at any trade timestamp.

Part 10 — Performance Metrics

Metric definitions as computed by the Lab (lab/backtest.py::_summarize):

- **Total return %** = net realised P&L / \$1,200 starting book × 100.
- **Net P&L** = Σ per-trade P&L (net of taker fee + slippage on both legs).
- **Win rate** = winning trades / total trades.
- **Profit factor** = gross wins / gross losses.
- **Sharpe (per-trade)** = mean(trade return %) / stdev; **Sortino** uses downside deviation only. Annualisation-free (per-trade), so comparable across timeframes.
- **Max drawdown %** = largest peak-to-trough drop of the realised-P&L equity curve (trades ordered by exit).
- **Expectancy** = net P&L / trades. **MFE/MAE** = avg max favourable / adverse excursion %. **Profit left on table** = MFE\$ – captured\$. **Trade quality (0–100)** = 0.5·capture + 0.3·MAE-term + 0.2·speed.

Portfolio-level headline (this run) — three independent \$1,200 books:

Metric	BTC/USD	SOL/USD	ETH/USD	Aggregate*
Total return %	-13.586	-23.583	-18.439	-18.54
Net P&L \$	-163.033	-282.994	-221.263	-667.29
Trades	197	249	227	673
Win rate %	34.0	36.5	37.0	—
Profit factor	0.33	0.25	0.30	—
Sharpe (per-trade)	-0.149	-0.289	-0.217	—
Sortino (per-trade)	-0.168	-0.268	-0.219	—
Max drawdown %	1.03	1.53	1.28	—
Avg MFE %	1.934	2.325	2.291	—
Avg MAE %	-1.694	-2.232	-2.036	—
Total profit left \$	448.85	717.26	611.35	1777.46
Avg trade quality	53.3	51.3	52.3	—

*Aggregate = summed across the three independent \$1,200 books (return = Σ netP&L / \$3,600). The report itself does not present a consolidated portfolio view; win-rate/PF/Sharpe are intentionally left blank at aggregate to avoid a misleading blend.

Part 11 — Strategy Performance Breakdown

UNAVAILABLE at the per-STRATEGY level. The supplied report aggregates all entries per SYMBOL and breaks results down by REGIME and by EXIT MODULE — not by strategy. Because only the three core models generated entries, per-strategy attribution exists in the run's stored data (strategy_breakdown bucket) but was not rendered in this PDF, so it cannot be quoted here without re-reading the production database. The closest available attributions (regime & exit module) are reproduced below.

BTC/USD — regime breakdown

Regime	N	Win %	Net P&L \$
COMPRESSION	10	40.0	-2.29
NEUTRAL	70	35.7	-54.67
RANGE	1	0.0	-1.75
REVERSAL	8	50.0	-6.02
TREND_UP	108	31.5	-98.31
Exit module	N	Win %	Net P&L \$
ATR	195	34.4	-162.58
EOD	2	0.0	-0.45

Verdict: UNDERPERFORMING - -13.6% at 34.0% win rate; the edge is weak in this window. Re-test other timeframes or parameter presets.

Best exit engine (return/drawdown): Fixed \$3.00 / \$2.25 - -7.45% net return at 34.0% win rate, profit factor 0.69, 0.80% max drawdown over 197 trades.

SOL/USD — regime breakdown

Regime	N	Win %	Net P&L \$
COMPRESSION	6	16.7	-7.69
NEUTRAL	58	41.4	-45.84
RANGE	5	40.0	-6.44
REVERSAL	6	0.0	-17.9
TREND_UP	174	36.8	-205.12
Exit module	N	Win %	Net P&L \$
ATR	249	36.5	-282.99

Verdict: UNDERPERFORMING - -23.6% at 36.5% win rate; the edge is weak in this window. Re-test other timeframes or parameter presets.

Best exit engine (return/drawdown): Fixed \$5.00 / \$4.00 - -25.25% net return at 30.9% win rate, profit factor 0.56, 1.89% max drawdown over 249 trades.

ETH/USD — regime breakdown

Regime	N	Win %	Net P&L \$
COMPRESSION	6	33.3	-4.59
NEUTRAL	61	34.4	-60.15
RANGE	5	40.0	-3.11
REVERSAL	8	25.0	-10.72
TREND_UP	147	38.8	-142.69
Exit module	N	Win %	Net P&L \$
ATR	225	36.9	-221.26
EOD	2	50.0	-0.01

Verdict: UNDERPERFORMING - -18.4% at 37.0% win rate; the edge is weak in this window. Re-test other timeframes or parameter presets.

Best exit engine (return/drawdown): Fixed \$5.00 / \$4.00 - -12.86% net return at 37.0% win rate, profit factor 0.73, 1.52% max drawdown over 227 trades.

Multi-timeframe comparison

BTC/USD

TF	Trades	Return %	Win %	MaxDD %	Avg MFE %	Avg MAE %
15m	0	—	—	—	—	—
30m	0	—	—	—	—	—
1h	197	-13.586	34.0	1.03	1.934	-1.694

SOL/USD

TF	Trades	Return %	Win %	MaxDD %	Avg MFE %	Avg MAE %
15m	0	—	—	—	—	—
30m	0	—	—	—	—	—
1h	249	-23.583	36.5	1.53	2.325	-2.232

ETH/USD

TF	Trades	Return %	Win %	MaxDD %	Avg MFE %	Avg MAE %
15m	0	—	—	—	—	—
30m	0	—	—	—	—	—
1h	227	-18.439	37.0	1.28	2.291	-2.036

Only the 1h series produced trades; 15m/30m produced none in this window — the models' edge is defined on 1h candles.

Exit-engine A/B/C comparison (identical entries, exit is the only variable)

BTC/USD

Exit config	PF	Win %	Expectancy \$	Net ret %	MaxDD %
Fixed \$2.00/\$1.50	0.34	20.3	-0.78	-12.75	0.98
Fixed \$3.00/\$2.25 (best)	0.69	34.0	-0.45	-7.45	0.8
Fixed \$4.00/\$3.00	0.68	35.0	-0.62	-10.2	0.97
Fixed \$5.00/\$4.00	0.64	36.0	-0.91	-15.01	1.32
ATR Baseline (x2.5, 14p)	0.33	34.0	-0.83	-13.59	1.03

SOL/USD

Exit config	PF	Win %	Expectancy \$	Net ret %	MaxDD %
Fixed \$2.00/\$1.50	0.42	24.1	-0.66	-13.62	0.87
Fixed \$3.00/\$2.25	0.51	27.7	-0.8	-16.5	1.06
Fixed \$4.00/\$3.00	0.52	28.1	-1.03	-21.42	1.39
Fixed \$5.00/\$4.00 (best)	0.56	30.9	-1.22	-25.25	1.89
ATR Baseline (x2.5, 14p)	0.25	36.5	-1.14	-23.58	1.53

ETH/USD

Exit config	PF	Win %	Expectancy \$	Net ret %	MaxDD %
Fixed \$2.00/\$1.50	0.37	21.6	-0.74	-14.04	0.97
Fixed \$3.00/\$2.25	0.49	26.9	-0.84	-15.94	1.18
Fixed \$4.00/\$3.00	0.63	32.2	-0.75	-14.25	1.38
Fixed \$5.00/\$4.00 (best)	0.73	37.0	-0.68	-12.86	1.52
ATR Baseline (x2.5, 14p)	0.3	37.0	-0.97	-18.44	1.28

Part 12 — Historical Trade Log

The complete trade log — **673 trades** (BTC/USD 197 · SOL/USD 249 · ETH/USD 227) — is provided in **ananta_trade_log.csv**, parsed directly from the report. Columns: symbol, trade #, entry/exit UTC, entry/exit price, size, net P&L \$, exit module.

Fields requested by the review that are NOT in the report's trade table (strategy, direction=all-long, SL/TP levels, indicators at entry/exit, signal confidence, entry/exit reasoning beyond the exit module, market regime per trade, news/event) are UNAVAILABLE from this run's PDF. The per-candle indicators CSV provides the full indicator state at every timestamp so any trade can be validated by joining on time.

Part 13 — Market Regime Detection

`classify_regime()` (regime.py, pure compute) labels each bar from the asset's own trailing window using EMA20/50/200, RSI14, ADX14, ATR14 (+percentile), Bollinger bandwidth (+percentile) and crude swing structure.

Regime	Definition
COMPRESSION	(BBwidth pct $\leq$ 30 AND ATR pct $\leq$ 35) OR ATR pct $\leq$ 10 — volatility collapse
REVERSAL	panic (RSI $\leq$ 22 & ATR pct $\geq$ 70) OR (RSI $\leq$ 32 & down-stack/lower-low)
TREND_UP	EMA20>50>200 AND price>EMA20 AND ADX $\geq$ 20 AND higher-high/higher-low
TREND_DOWN	EMA20<50<200 AND ADX $\geq$ 20
RANGE	ADX < 20
NEUTRAL	none of the above

Priority when several conditions hold: compression > reversal > strong-trend > range > neutral. **Strategy switching already exists** at the ROUTER level (regime admits which model may act), but there is NO dynamic parameter/allocation rotation per regime — that is a clear integration point (Part 16/17). Fake-breakout, accumulation/distribution and explicit capitulation labels are not separately modelled.

Part 14 — AI Components

- **Where AI is used:** (1) Macro-context engine (Gemini 3.1 Pro via the Emergent key) turns a news summary into a structured BIAS/CONFIDENCE/REASON, MD5-cached to avoid re-spend; (2) 'Ask Ananta' copilot; (3) weekly performance review.
- **Where decisions are deterministic:** ALL entries and exits, regime, routing, sizing and risk are pure compute (zero LLM). The AI cannot create, rescue or block a trade — `risk_engine.fuse_signals` is technical-first by design.
- **Limitations:** macro bias is advisory context only; no reinforcement learning; no LLM in the hot path.
- **Pipeline:** deterministic reasoning is recorded per decision (reason codes, rejections) for the analytics/research layer.

Part 15 — Current Limitations (real-world performance risks)

- **Negative edge in this window:** all three symbols lost money (-13.6% / -23.6% / -18.4%) with profit factors 0.25–0.33 and 34–37% win rate — the current entry+ATR-exit combination has no demonstrated edge on 1y/1h.
- **Only 3 of 11 strategies actually traded** in the backtest — the eight declarative strategies were selected but not wired into the offline entry scan, so the report does NOT validate them.
- **Exit gives back gains:** avg profit-left-on-table \$2.28–\$2.88/trade (\$448–\$717 per book) — MFE $\sim +1.9\text{--}2.3\%$ but captured returns are negative; the ATR trail exits late / whipsaws (PF 0.25–0.33 vs Fixed-\$ 0.56–0.73 best).
- **Execution realism:** 0.05%/leg slippage & taker fees are modelled, but exchange latency, partial fills, orderbook depth, funding and downtime are not.
- **Whipsaw / sideways:** NEUTRAL + TREND_UP regimes dominate trades and are the biggest losers (BTC TREND_UP $-\$98.31$ over 108 trades) — the models mis-fire in chop and shallow trends.
- **Correlation:** BTC/ETH/SOL are highly correlated; three parallel books overstate diversification.
- **Overfitting/robustness:** deterministic (good) but a single 1y window on 3 assets is thin; no walk-forward or Monte-Carlo in THIS run (the Lab supports both separately).
- **Indicator lag & parameter sensitivity:** EMA/ATR/RSI are lagging; the exit-comparison shows results swing materially with the exit parameter — high sensitivity.

Part 16 — Planned Improvements (roadmap)

- Wire the 8 declarative strategies into the offline entry scan so the Lab can validate them (prerequisite to any multi-strategy claim).
- Regime-aware strategy rotation & dynamic parameter selection (only trade a model in the regimes where its expectancy is positive).
- Dynamic / volatility-scaled position sizing (replace flat \$75); risk-per-trade in R.
- Adaptive exits: the A/B/C test already shows Fixed-\$ beats the ATR trail here — tune stop/trail per regime; consider partial profit-taking earlier to capture the \$448–\$717 left on the table.
- Walk-forward optimisation + Monte-Carlo (both exist in the Lab) as gating before promotion; Bayesian parameter search.
- Portfolio construction with correlation filters across BTC/ETH/SOL; news/volatility filters; orderbook-depth aware execution.
- Longer-term: meta strategy selection and reinforcement learning (research track).

Part 17 — Recommendations for the External Quantitative Reviewer

Focus areas, ranked by expected impact on this specific result set:

- **Exit optimisation first (highest ROI):** the ATR trail is the single worst config on 2 of 3 symbols (PF 0.25–0.33). Fixed-\$ TP/SL already lifts BTC to PF 0.69 and ETH to 0.73 on identical entries. Grid/Bayesian-tune the exit per regime and re-run.
- **Entry quality & regime gating:** TREND_UP and NEUTRAL produce the most trades and the biggest losses — tighten or disable core models in those regimes; the win rate (34–37%) needs either a higher bar or an asymmetric payoff.
- **Validate the 8 declarative strategies:** they are untested by this run; wire them in before any multi-strategy claim.
- **Reduce late entries / premature exits:** MFE $\approx$ +1.9–2.3% shows the move exists; capture is the problem — earlier partials + smarter trailing.
- **Robustness:** run walk-forward + Monte-Carlo across more assets/windows; add correlation-aware portfolio sizing.
- **Parameter sensitivity:** quantify sensitivity (the exit sweep already hints it is high) and prefer robust plateaus over sharp optima.
- **Data to request for a deeper audit:** per-trade strategy tag + indicators-at-entry (available in the engine's trade dicts / production DB), and the equity-curve series per book.

Companion artifacts (this package): **ananta_trade_log.csv** (every trade), **ananta_per_candle_indicators.csv** (25,995 rows — full indicator series with engine formulas over the identical OHLCV window), **ananta_engine_config.json** (complete machine-readable configuration). Together with the original backtest PDF they let a reviewer independently validate every trade and recommend evidence-based parameter changes.